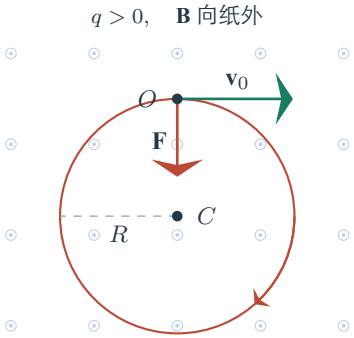
M5-171 由正则动量恢复洛伦兹力分量



$$L = \tfrac{1}{2}m\mathbf{v}^2 - qBy\dot{x}$$

$$\partial L/\partial x = 0$$

$$\frac{d}{dt}(mv_x - qBy) = 0$$